

CSCE 633 MACHINE LEARNING · MODULE 05 · LECTURE NOTES

Evaluation and Metrics

Choosing the number you will be judged by.

MODULE OUTCOMES

Read a confusion matrix and derive the metrics from it.

Choose between precision, recall, F1, and the curves.

Explain why ROC misleads under heavy imbalance.

Measure and interpret calibration.

Choose a metric from the costs of the errors.

1 The confusion matrix

		PREDICTED		
		positive	negative	
ACTUAL	pos	TP	FN	<- FN: the MISS
	neg	FP	TN	<- FP: the FALSE ALARM

precision = $TP / (TP + FP)$ "of what I flagged, how much was real?"
 -> the cost of a FALSE ALARM

recall = $TP / (TP + FN)$ "of what was real, how much did I find?"
 -> the cost of a MISS

specificity = $TN / (TN + FP)$

F1 = harmonic mean of precision and recall

ACCURACY = $(TP + TN) / \text{everything}$
 -> meaningless under imbalance: predicting "negative" always scores 99% on a 99%-negative dataset, with recall of ZERO.

ACCURACY IS THE WRONG DEFAULT

Accuracy weights every error equally and implicitly assumes balanced classes. Both assumptions are false in most real problems, and neither is usually examined.

On a dataset that is 99% negative, a model that predicts 'negative' unconditionally scores 99% — and has found nothing, learned nothing, and is useless for the purpose it was built for. Its recall is exactly zero.

Yet accuracy remains the default metric nearly everywhere — the library default, the first number in the notebook, and the one that reaches a summary slide where its denominator is invisible.

So report the confusion matrix rather than a scalar, and when a scalar is genuinely required, **choose it from the costs of the two error types** (Module 01 §3) rather than from convention. **Which metric you optimise is a statement about which error you are willing to make**, and making that statement explicitly is the point of this module.

2 Curves

ROC IS OPTIMISTIC UNDER HEAVY IMBALANCE; USE PR

The ROC curve plots the true positive rate against the *false positive rate*, and $FPR = FP / (FP + TN)$ — its denominator is the total number of negatives.

So when negatives vastly outnumber positives, even a large absolute number of false positives produces a tiny false positive rate, and the ROC curve hugs the top-left corner while the model is, in practice, flagging mostly noise.

The precision–recall curve uses precision instead, whose denominator is the number of positive predictions you actually made — so those same false positives dominate it and the curve shows the problem plainly.

At 1% positive prevalence, a model with an ROC AUC of 0.95 can easily have a precision around 10% at a useful recall — nine false alarms for every real detection, which is the number the person reviewing the alerts experiences. **Report the PR curve for imbalanced problems**, and reserve ROC for when the classes are comparable in size.

Situation	Metric	Why
Balanced classes, equal error costs	Accuracy.	The one case in which it is honest.
Rare positives	Precision–recall curve; average precision.	ROC flatters and PR does not — see the callout.
Misses are expensive	Recall, reported at a stated precision floor.	Medical screening, safety systems, fraud detection — where a missed positive is the costly outcome.

Situation	Metric	Why
False alarms are expensive	Precision, at a stated recall floor.	Alerting, and anything a human must review — alert fatigue is a real failure mode (CSCE 606 Module 11).
The probabilities are used downstream	Log loss, Brier score, and a reliability diagram.	Discrimination alone is not enough — §3.
Ranking	NDCG, MAP, MRR.	The order matters more than any threshold (CSCE 670 covers these).
Regression	MAE against RMSE.	RMSE punishes large errors disproportionately and MAE does not — the same choice as Module 01 §3's loss discussion, now at evaluation time.

'X at a floor of Y' is usually the honest form of a metric — 'recall of 0.82 at precision 0.90' states an operating point you could actually deploy, whereas an aggregate like AUC averages over thresholds including many you would never use.

3 Calibration

DISCRIMINATION AND CALIBRATION ARE DIFFERENT PROPERTIES

Discrimination asks: does the model rank positives above negatives? This is what AUC measures, and it is invariant to any monotonic transformation of the scores.

Calibration asks: when the model says 0.7, are about 70% of those cases actually positive? This is what a reliability diagram measures.

A model can discriminate perfectly and be badly calibrated. Take a perfectly calibrated model and square all its probabilities: the ranking is unchanged, so AUC is identical, and every probability is now wrong. **AUC cannot see this at all.**

And if any downstream decision uses the probability as a number — an expected-value calculation, a cost-based threshold, combining with another model's output, or showing a confidence to a user — **calibration is the property you actually need, and the metric everyone reports does not measure it.**

- **Reliability diagram.** Bin the predictions by predicted probability, and plot the observed positive frequency in each bin against the mean predicted probability. **The diagonal is perfect calibration;** above it is underconfidence, below is overconfidence.
- **Brier score and log loss are *proper scoring rules*** — they are minimised by reporting the true probabilities, so they reward calibration and discrimination together and cannot be gamed by distorting confidence.
- **Platt scaling:** fit a one-dimensional logistic regression mapping the model's scores to probabilities. Simple, needs little data, and assumes a sigmoid-shaped distortion.
- **Isotonic regression:** fit a monotonic step function instead. More flexible, makes no shape assumption, and needs considerably more data or it overfits.
- **And fit the calibrator on data the model did not train on,** or you have calibrated to the training set and learned nothing. This is Module 04 §2's rule again — **calibration is a learned step and belongs inside the pipeline. Modern neural networks are badly calibrated and systematically overconfident,** which genuinely surprised the field when Guo and colleagues measured it, and older smaller networks were better.

4 Class imbalance

Approach	Effect	Caution
Move the decision threshold	Trades precision against recall along the curve.	Free, instant, and fully reversible. Try this first — see the callout.
Class weights in the loss	Reweights the contribution of each class.	Usually sufficient, and it invents nothing and discards nothing.

Approach	Effect	Caution
Undersample the majority	Balances the classes.	Discards real information , which is a strange thing to do when data is the scarce resource.
Oversample the minority	Duplicates minority rows, which encourages memorising them.	And it breaks cross-validation if done before splitting — copies of the same row land in both train and test.
SMOTE and relatives	Synthesises new minority points by interpolating between neighbours.	Invents data that was never observed, and the published evidence on properly-evaluated problems is genuinely mixed despite its popularity.
Collect more positives	The actual fix.	Usually expensive, and usually the right thing to ask for.

THRESHOLD FIRST, RESAMPLING LAST

A probabilistic model trained on imbalanced data is frequently perfectly fine. It has learned the conditional probabilities; it simply needs a decision threshold other than the 0.5 that encodes equal costs and equal priors.

So move the threshold and examine the precision–recall curve before touching the data at all. It costs one line, it is reversible, and it frequently resolves the problem entirely.

Then class weights, which reweight the loss function so that minority errors count more — without inventing any data or discarding any.

Resampling last, and strictly inside the cross-validation fold. A substantial portion of reported SMOTE improvement is leakage from resampling before splitting (Module 04 §2): synthetic points interpolated from a row that then lands in the test set produce spectacular and entirely fictitious scores.

When the evaluation is done correctly, the gains largely evaporate, which is a result worth knowing before adopting the technique.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Saito & Rehmsmeier — The Precision-Recall Plot Is More Informative than the ROC Plot (free)	The §2 argument , with worked examples at several prevalences. Short and conclusive.
Guo, Pleiss, Sun & Weinberger — On Calibration of Modern Neural Networks (free)	The §3 result. Modern networks are far less calibrated than older ones, measured and explained.

Resource	What it covers for this module
scikit-learn — metrics and probability calibration guides (free)	The implementations, with unusually good discussion of when each metric is appropriate.
Blagec et al.; and the imbalanced-learn documentation (free)	The \$4 resampling pitfalls , including the resample-before-split error, documented by the library that implements the methods.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Build a confusion matrix for your project model and derive all five metrics from it by hand.
- 2 **Build the 99%-negative demonstration:** report accuracy, precision, recall, and F1 for a model that always predicts negative.
- 3 Plot the ROC and PR curves for the same model at 50%, 10%, and 1% prevalence. **Report how differently they look.**
- 4 Compute AUC and average precision at each prevalence and explain the divergence.
- 5 **Plot a reliability diagram** for a logistic regression, a random forest, and a boosted model on the same data.
- 6 **Square every predicted probability** and show AUC is unchanged while calibration is destroyed.
- 7 Apply Platt scaling and isotonic regression and compare the reliability diagrams.
- 8 **Tune the threshold** on an imbalanced problem and report the precision–recall trade at several points.
- 9 **Oversample before splitting and after**, and report both scores. Explain the gap.
- 10 Choose your project's reporting metric and write one paragraph justifying it from the error costs.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Define precision, recall, specificity, and F1 from the confusion matrix.
- 2 Why is accuracy the wrong default?
- 3 Why does ROC flatter under heavy imbalance?
- 4 Give seven situations and the right metric for each.
- 5 Distinguish discrimination from calibration and give an example of high one and low the other.
- 6 Name two calibration methods and say where the calibrator must be fitted.
- 7 Give six responses to imbalance and the right ordering.
- 8 Why does resampling before splitting produce fake scores?

